Masteraid

Modular Mastering Console • Technical Reference Manual

DSP Foundations & Airwindows Algorithms: Masteraid incorporates open-source DSP algorithms developed by **Chris Johnson (Airwindows)**, including non-linear saturation, slew rate filtering, and discrete stage modeling. To support his independent work, visit **patreon.com/airwindows**.



Figure 0.1 — Masteraid Dual-Theme Precision Analog Mastering Console (Dark Studio & Smoked-Glass Light Modes)

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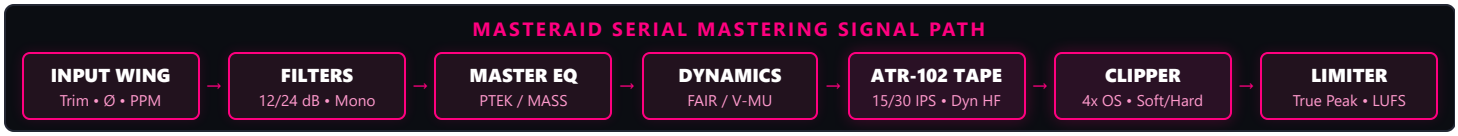
01 Console Architecture & Serial Signal Flow

Masteraid is a modular mastering console that brings together vacuum-tube and discrete transformer models, tape machine emulation, two switchable equalizers, two switchable variable-mu compressors, oversampled clipping, and a true-peak limiter in a 64-bit floating-point processing chain.



Figure 1.1 — Masteraid Full Rack Console Interface (Hardware Dark Mode)

The signal flows in serial mastering order: input gain staging and corrective filtering come first, followed by equalization and compression, then tape saturation, peak clipping, and final true-peak limiting.



Why this specific order matters: Equalizing before dynamics ensures the compressor sidechain reacts to the intended tonal balance rather than uncontrolled low-frequency peaks. Magnetic tape saturation adds progressive non-linear compression, and the clipper catches fast transient spikes so the limiter does not have to work as hard.

02 Left Wing: Input Stage & Master Filtering Controls

The left wing handles gain staging, bandwidth preparation, and stereo mono-compatibility auditing prior to downstream equalization, tape saturation, and dynamic control:



Figure 2.1 — Input Stage, Trim, PPM/RMS Ladder & Filter Network

PARAMETER	RANGE / OPTIONS	DEFAULT	ENGINEERING FUNCTION
IN (Input Bypass)	Engaged / Bypassed	Engaged	Section bypass toggle. When disengaged, trim and filtering are bypassed while input peak/RMS meters remain fully active.
TRIM	-10.0 dB to +10.0 dB	0.0 dB	Stepped input gain in 0.5 dB increments for fine gain staging into downstream analog stages.
PHASE (Ø)	Normal / Inverted (180°)	Normal	Inverts absolute polarity of the stereo bus to test phase correlation against references.
MONO	Stereo / Mono Sum	Stereo	Sums left and right channels to mono for quick phase-cancellation and compatibility checks.
LO CUT (Highpass)	20, 30, 40, 60, 80 Hz • 12 / 24 dB/oct	OFF	Linkwitz-Riley highpass filter. Selectable 12 dB/oct (2nd order) or 24 dB/oct (4th order). Cleans up infrasonic rumble.
HI CUT (Lowpass)	14k, 16k, 18k, 20k, 24k Hz • 12 / 24 dB/oct	OFF	Linkwitz-Riley lowpass filter. Selectable 12 dB/oct or 24 dB/oct slopes. Limits ultrasonic content before distortion stages.
BASS MONO & AMT	OFF, 40, 60, 80, 120, 160, 200 Hz • 0..100%	OFF (70%)	Phase-aligned elliptical crossover (12 or 24 dB/oct) that narrows low frequencies below the corner frequency to mono.
INPUT METERS	-60.0 dBFS to +12.0 dBFS	Live	Dual bar meters displaying PPM instantaneous peak hold (0.7 s release) and 300 ms integrated RMS level.

03 Parametric Equalizer: PTEK Tube vs MASS Parametric

The equalizer stage features two distinct circuit topologies, selectable via the illuminated top model selector:

> **Model Selection:** Switch between PTEK and MASS at any time. Filter states crossfade smoothly without clicks.

PTEK Mode: Passive Program Tube Equalizer

PTEK models a passive LC inductor-capacitor filter network with vacuum tube makeup amplification based on the classic EQP-1A architecture:



Figure 3.1 — PTEK Mode: Passive LC Filter Topology & Program Tube Gain Makeup

- **Low-End Interaction (Simultaneous Boost & Attenuation)**: Because the boost and attenuation circuits use different corner frequencies and slopes, applying boost and cut together creates a resonant low-frequency shelf with a gentle dip near 250–400 Hz, clearing muddy lower-mid buildup while tightening the sub-bass.
- **Low Frequency Selectors**: Stepped selection at 20, 30, 40, 60, 100, and 150 Hz with continuous Boost (0..10 = +13.5 dB) and Attenuate (0..10 = -17.5 dB) controls.
- **High Frequency Boost & Attenuation**: High boost provides 7 selectable center frequencies (3k, 4k, 5k, 8k, 10k, 12k, 16k Hz) and variable bandwidth (Q: 0 to 10). A separate high-frequency shelf attenuator operates at 5k, 10k, or 20k Hz (0 to -16.0 dB).

MASS Mode: Four-Band Parallel Parametric Mastering Equalizer

MASS is a 4-band parallel parametric mastering equalizer. Unlike serial biquad designs where phase shifts accumulate stage by stage, MASS uses a parallel topology where all four bands process the input independently and sum with the dry signal, maintaining consistent transient response and stereo correlation.



Figure 3.2 — MASS Mode: Four-Band Parallel Inductor Mastering Equalizer with Proportional-Q

BAND	STEPPED FREQUENCIES	GAIN RANGE	Q BANDWIDTH	MASTERING FOCUS & APPLICATION
LOW	33, 47, 68, 100, 150, 220 Hz	-9.0 to +9.0 dB	0.4 to 5.0	Sub-bass foundation, kick drum weighting, and infrasonic cleanup.
LO MID (LMF)	150, 220, 330, 470, 680, 1k Hz	-9.0 to +9.0 dB	0.4 to 5.0	Lower-midrange density, snare body, acoustic instrument resonance.
HI MID (HMF)	680, 1k, 1.5k, 2.2k, 3.3k, 4.7k Hz	-9.0 to +9.0 dB	0.4 to 5.0	Vocal articulation, snare attack, presence, and sibilance control.
HIGH (HF)	3.3k, 4.7k, 6.8k, 10k, 15k, 20k Hz	-9.0 to +9.0 dB	0.4 to 5.0	High-frequency air, cymbal extension, and top-end opening.

Proportional-Q Characteristic Curve

Filter bandwidth adjusts proportionally to gain. Subtle adjustments (±0.5 to ±2.0 dB) stay broad for gentle tonal shaping. As boost or cut increases toward ±9.0 dB, bandwidth progressively narrows for targeted correction without altering neighboring frequencies.

Hardware Controls & Band Bypass

- **Frequency Collar**: 6-position stepped selector locking onto inductor frequency taps.
- **Gain Rotor**: ±9.0 dB continuous gain control with center detent at 0.0 dB.
- **Q Trim Knob**: Adjusts filter bandwidth from broad (0.4) to narrow (5.0).
- **Shelf / Bell Toggle**: Switches between **SHELF** (UP) and **BELL** curve (DOWN). In Shelf mode, the Q knob sets slope steepness.
- **Band Bypass**: Click the frequency/gain readout above each band to toggle it ON or OFF.

04 Mastering Dynamics: FAIR vs V-MU

The dynamics section provides variable-mu compression with program-dependent attack and recovery curves, providing cohesive program leveling across stereo stems.

FAIR Mode: Vintage Fairchild 670 Variable-Mu

FAIR mode emulates a push-pull variable-mu gain reduction circuit with feedback detection. It incorporates six standardized time constant presets (TC 1 through TC 6) with multi-slope recovery on settings 5 and 6, and adaptive release on low frequencies to prevent waveform distortion:



Figure 4.1 — FAIR Mode: Push-Pull Variable-Mu Topology with Calibrated Time Constants

TIME CONSTANT	ATTACK TIME	RECOVERY (RELEASE) CHARACTERISTIC	RECOMMENDED APPLICATION
TC 1	0.2 ms	0.3 seconds	Fast transient control, aggressive drum and rhythmic punch.
TC 2	0.2 ms	0.8 seconds	Standard pop and rock bus leveling; general program cohesion.
TC 3	0.4 ms	2.0 seconds	Slower, flowing compression for acoustic, jazz, and vocal arrangements.
TC 4	0.8 ms	5.0 seconds	Macro-dynamic leveling for classical and wide-dynamic mixes.
TC 5 (Auto)	0.4 ms	Dual-stage automatic: 2.0s for short peaks, 10.0s for sustained signals.	Dynamic material with high-energy sections; prevents pumping.
TC 6 (Auto)	0.2 ms	Tri-stage automatic: 0.8s fast peaks, 3.0s medium, 25.0s sustained passage.	Full album mastering; transparent program-dependent leveling.

V-MU Mode: Modern Variable-Mu Mastering Compressor

V-MU mode provides independent stepped control over envelope timing, compression ratio, and sidechain frequency weighting:



Figure 4.2 — V-MU Mode: Stepped Envelope Timing, SC HPF & Parallel Wet/Dry Mix

- **Stepped Timing & Ratio**: Stepped Attack (Fast, M-Fast, Med, M-Slow, Slow), Recovery (Fast, Med, Slow, Auto), and Compress / Limit mode switch.
- **Sidechain Highpass Filter (SC HPF)**: Stepped 80 Hz / 100 Hz filter prevents heavy low frequencies from driving unnecessary gain reduction.
- **Parallel MIX & Automatic Makeup**: 0% to 100% wet/dry blend for parallel compression, plus AUTO makeup gain compensation.
- **Stereo Gain Reduction Meters**: Left and right channel gain reduction displays responding with standard 300 ms integration time.

05 Analog Coloration Cassettes

Both the EQ and Dynamics sections feature an independent **Hardware Coloration Cassette** on their output stages, modeling discrete active amplifier topologies and output transformer magnetic saturation:

7 Active Amplifier Topologies

1. **BYPASS**: Linear signal transmission without added harmonic coloration.
2. **TRIODE (12AX7)**: Single-ended triode circuit producing predominantly second-order harmonic distortion.
3. **PUSH-PULL TUBE**: Balanced push-pull circuit where even-order harmonics cancel, leaving third-order harmonics.
4. **PENTODE (EF86)**: High-gain pentode stage with extended upper-frequency harmonic distribution.
5. **FET (1176 Style)**: Field-effect transistor circuit with fast transient snap and odd-order harmonics under drive.
6. **MOSFET**: Solid-state circuit providing soft-clipping saturation characteristics with low intermodulation.
7. **GERMANIUM DIODE**: Non-linear diode conduction profile with progressive low-level compression and soft high-frequency rounding.



Figure 5.1 — Hardware Coloration Cassette Face

5 Transformer Core Types

- **BYPASS**: Transformerless electronic output.
- **STEEL CORE**: Silicon steel lamination producing low-frequency harmonic distortion and magnetic hysteresis.
- **HIGH-NICKEL CORE**: Nickel alloy core offering low phase distortion and wide frequency linearity.
- **AMORPHOUS ALLOY**: Non-crystalline magnetic core with fast transient response and high-frequency extension beyond 30 kHz.
- **MU-METAL**: High-permeability nickel-iron shielding alloy providing flat response and subtle low-level magnetic saturation.

Coloration Controls & Operation (PUSH, HOLD AUTO & LEARN):

- **PUSH (Drive Tiers)**: Direct click cycles between three calibrated saturation stages: **Subtle** (nominal operating point), **Pushed** (+4.5 dB core drive with rich harmonic density), and **Smashed** (+11.5 dB extreme drive pushing tubes and transformers into aggressive non-linear saturation).
- **AUTO (Level Neutrality)**: Click to toggle automatic level compensation ON or OFF using the latched trim values.
- **Hold AUTO (Learn Mode)**: Press and hold the AUTO button while playing program material. Masteraid runs three background shadow stages to measure the exact power ratio across Subtle, Pushed, and Smashed tiers simultaneously. The resulting level trims are permanently latched per amplifier/transformer pairing.
- **Double-Click AUTO**: Clears and resets the stored learned calibration trim for the current amplifier and transformer model pairing.
- **MATCH**: Instantly locks the period-correct active amplifier and transformer topologies associated with the active EQ or compressor circuit.

06 Top Deck: ATR-102 Mastering Magnetic Tape Machine

The tape stage models 1/2-inch two-track magnetic mastering tape machines, including magnetic oxide saturation, ultrasonic record bias, low-frequency head contour resonance, and transport wow/flutter:

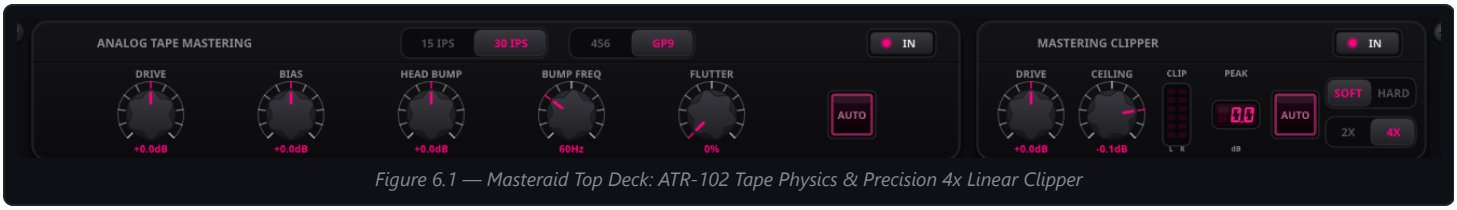


Figure 6.1 — Masteraid Top Deck: ATR-102 Tape Physics & Precision 4x Linear Clipper

- **Dynamic High-Frequency Compander**: Implements high-frequency pre-emphasis conditioning prior to tape saturation and complementary decoding on playback, reducing ultrasonic intermodulation and preventing high-frequency loss during heavy saturation.
- **15 IPS vs 30 IPS Tape Speeds**: 15 IPS (Inches Per Second) emphasizes lower-frequency head bump resonance and low-mid tape compression; 30 IPS delivers extended high-frequency linearity beyond 28 kHz with lower modulation noise.
- **456 vs GP9 Tape Formulations**: 456 (+6 dB operating level) produces standard magnetic tape saturation that rounds peak transients. GP9 (+9 dB operating level) provides higher headroom and lower harmonic distortion.
- **DRIVE (-6 to +12 dB) & BIAS (-10 to +10 dB)**: Drive sets input signal level relative to the magnetic tape saturation curve. Bias adjusts the AC ultrasonic record bias current and high-frequency slew rate damping.
- **HEAD BUMP (0 to 6 dB) & BUMP FREQ (30 to 150 Hz)**: Models the physical repro head contour effect. Adjusts the low-frequency resonant peak frequency (tunable from 30 to 150 Hz) with an active highpass filter to control sub-audible DC buildup.
- **FLUTTER (0.0% to 100.0%)**: Simulates mechanical transport irregularities and capstan flutter using uncorrelated stereo modulation.
- **AUTO (Level Neutrality)**: Level-matched saturation compensation counterbalancing input drive in real time to facilitate objective timbre evaluation.

07 Top Deck: Mastering Clipper (Noise-Masked Soft Ceiling)

The mastering clipper applies 4x linear-phase oversampling to control peak crest factor prior to the limiter stage:

- **Soft Ceiling**: Applies smooth knee transition shaping to eliminate hard discontinuity artifacts, reducing peak levels with minimal disruption to the overall dynamic envelope.
- **Soft vs Hard Shape**: Soft knee progressively rounds transient peaks with odd-order harmonics; Hard knee clips strictly at the ceiling threshold with zero time-constant envelope alteration.
- **DRIVE (-6.0 to +12.0 dB) & CEILING (-6.0 to 0.0 dBTP)**: Drive sets signal level into the clipping stage; Ceiling defines the maximum peak output.
- **AUTO (Clipper)**: Automatically compensates for drive gain so clipping intensity can be evaluated without shifts in perceived loudness.
- **Fast Metering**: Peak indicator displaying instantaneous peak attenuation in decibels per channel.

08 Right Wing: Brickwall Limiter & Broadcast Metering

The final stage combines a lookahead true-peak brickwall limiter with inter-sample peak detection and ITU-R BS.1770-4 / EBU R128 loudness metering:



Figure 8.1 — True-Peak Lookahead Limiter & EBU R128 Loudness Display

CONTROL / METER	RANGE / CALIBRATION	ENGINEERING FUNCTION
IN (Limiter Bypass)	Engaged / Bypassed	Engages or bypasses the limiting stage while loudness meters remain operational.
GAIN	0.0 dB to +18.0 dB	Limiter input drive pushing program material toward the ceiling.
CEILING	-6.0 dBTP to 0.0 dBTP	Sets the absolute True Peak output ceiling (standard recommendation -0.1 to -1.0 dBTP).
LOOKAHEAD	0.0 ms to 2.0 ms	Internal lookahead buffer allowing the limiter envelope to anticipate transient peaks without overshoot.
RELEASE & AUTO	10 ms to 1000 ms • Auto ON/OFF	Envelope recovery time. When Auto is engaged, release time dynamically lengthens on low-frequency content to prevent waveform distortion.
SMOOTH PUNCH	ON / OFF	Applies a lookahead envelope modifier that preserves initial transient edges during gain reduction.
LIFT	0.0% to 100.0%	Upward dynamic processor that raises low-level program content without increasing peak output ceiling.
TRUE PK (True Peak)	ON / OFF	Engages 4x oversampled inter-sample peak detection to prevent D/A inter-sample clipping.
UNITY	ON / OFF	Inverts limiter gain for gain-matched A/B comparison without volume bias.
LUFS METERS	Momentary, Short-Term, Integrated	EBU R128 compliant loudness metering. Alt+Click on the readout to reset integrated measurement.

09 Preset Management & System Directory Paths

Masteraid stores presets as standalone `.mqnpreset` files in the standard operating system documents directory for instant cross-platform exchange:

> **Standard User Preset Locations:**
> - **Windows**: `%USERPROFILE%\Documents\MQNdsp\Masteraid\Presets\`
> - **macOS**: `~/Documents/MQNdsp/Masteraid/Presets/`
> - **Linux**: `~/Documents/MQNdsp/Masteraid/Presets/` (or `$XDG_DOCUMENTS_DIR/MQNdsp/Masteraid/Presets/`)

Click the center preset display to browse the preset drawer. Click **[OPEN FOLDER]** to open the OS file manager directly. Presets placed in this directory become available immediately without restarting the host DAW.

10 Dual Themes, A/B Comparison & Shortcuts Reference



Figure 10.1 — Dual Visual Schemes: Dark Studio Finish (Left) vs Light Smoked-Glass Aesthetic (Right)

Visual Themes: Click ☀ (Sun) or ☾ (Moon) in the top header to toggle between Dark Mode (designed for low-light control rooms) and Light Mode (high-contrast daylight aesthetic).
A/B Matrix: The lower-right illuminated **A** and **B** buttons allow instant auditioning between two independent mastering configurations with zero audio interruption.

Mouse & Keyboard Shortcuts Reference

ACTION / SHORTCUT	TARGET CONTROL	OPERATIONAL BEHAVIOR
Ctrl + Drag (Cmd on macOS)	Stepped or Continuous Knobs	**Fine Precision**: Bypasses stepped detents for ultra-smooth fine-tuning (0.0022 precision).
Alt + Click (Option on macOS)	Any Knob, Slider, or Switch	**Reset Default**: Restores the selected parameter to its calibrated default mastering value.
Double-Click	Any Knob, Slider, or Switch	**Quick Reset**: Alternate one-handed shortcut to return any control to default.
Vertical Drag	Stepped Knobs & Selectors	**Detent Snap**: Snaps through calibrated inductor taps, Q bandwidths, or time constants.
Mouse Wheel	Any Knob or Slider	**Fine Step**: Increments or decrements detents and values with tactile resolution.
Direct Click	Buttons & Model Selectors	Toggles bypass states, switches circuit models (PTEK/MASS, FAIR/V-MU), or cycles saturation tiers.
Alt + Click on LUFS	LUFS Integrated / LRA Box	**Reset Loudness**: Resets EBU R128 integrated LUFS and LRA calculations for a new pass.
Book Icon (📖) Button	Top Navigation Header	**Open Manual**: Launches this technical reference guide in your system PDF viewer.

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